

WEEK
4

Boreal and Great Lakes Transition

Instruction | retrieval | multi-organ practice | field-ready reference

Public learner edition

This packet contains the teaching guide, cumulative reference, and formative practice for the week. Secure graded assessments remain in the online course so unfamiliar examination specimens and answer material are not published.

Student / participant

Date started

Planned pace / study notes

Field standard

Identify the narrowest rank the visible evidence can defend. A common name OR a scientific name satisfies an ordinary identity prompt unless the activity explicitly tests nomenclature. State uncertainty instead of converting weak evidence into a confident species claim.

Use this packet alongside the online course at cochetopa.co/course/. Work the instruction first, complete retrieval without looking back, and use the secure online assessment when ready for grading.

Week 4 overview

Learning objectives

1. Identify boreal-transition *Populus*, alder, and mountain-ash taxa from organs and whole-tree form.
2. Transfer earlier hardwood and conifer identifications into western Great Lakes and southern boreal contexts.
3. Distinguish morphology-supported identification from context-driven guessing.
4. Retrieve names and silvics cumulatively through four modules.
5. Complete the first substantial cumulative practical and field-context assessment.

Species introduced this week

Common name	Scientific name	Family	Tier
Quaking aspen	<i>Populus tremuloides</i>	Salicaceae	A
Bigtooth aspen	<i>Populus grandidentata</i>	Salicaceae	A
Balsam poplar	<i>Populus balsamifera</i>	Salicaceae	B
Speckled alder	<i>Alnus incana</i> subsp. <i>rugosa</i>	Betulaceae	B
American mountain-ash	<i>Sorbus americana</i>	Rosaceae	B
Showy mountain-ash	<i>Sorbus decora</i>	Rosaceae	B

Cumulative review

40 taxa remain eligible. Earlier species do not disappear from retrieval merely because the weekly theme changes.

Confusion sets

Sugar / black / Norway maple: Petiole latex; leaf underside hair and stipules; bud size/shape; samara angle

Request: Fresh-broken petiole; lower surface; terminal bud; ripe samara

Striped / mountain maple: Green-white striped bark versus unstriped; leaf size/base; fruit cluster posture; growth form

Request: Young stem; intact leaf; fruit cluster; whole plant

Paper / heart-leaved / gray birch: Leaf base and vein count; twig resin glands; peeling versus tight bark; black branch marks; elevation

Request: Several mature leaves; resin-warty twig; bark sheet; whole growth form; location

Yellow / sweet birch: Both wintergreen; yellow birch golden curling bark and spur shoots versus sweet birch dark blocky bark

Request: Scratched twig; young and mature bark; fruiting strobile; lower leaf surface

White / black / red / Norway spruce: Cone size and scale margin; twig hairs; bud resin; foliage color/odor; pendulous branchlets

Request: Full cone; twig macro; bud; whole crown; regional site

Spruce / fir / hemlock: Woody pegs versus pad base versus petiole; needle cross-section; cone orientation and persistence

Request: Needle attachment macro; underside; terminal shoot; intact or attached cone

Red / jack / Scots / pitch pine: Fascicle count/length; bend test; cone curvature/prickles; upper-bole orange bark; epicormic shoots

Request: Several complete fascicles; cone and umbo; upper and lower bark; whole crown

Tamarack / European larch: Cone size and scale count/edge; needle cluster density; native peatland versus planting

Request: Full cone; spur shoot; cluster count; site/planting context

Black / pin / chokecherry: Fruit cluster architecture; petiole glands; lower-midvein hair; bark age; whole pioneer/clonal form

Request: Raceme or cluster; petiole macro; lower leaf; mature bark; whole plant

American hornbeam / eastern hophornbeam: Muscular smooth bole and three-lobed fruit bracts versus shreddy bark and hoplike sacs

Request: Bole; fruit; bud; wet versus dry site

Yellow birch / black cherry bark: Golden horizontal curls and wintergreen twig versus burnt-chip plates, petiole glands, and almond twig odor

Request: Young and old bark; scratched twig; leaf petiole and underside; fruit

Serviceberry / mountain-ashes / chokecherry: Simple versus compound leaves; pome calyx versus drupe; leaflet proportions; buds; fruit architecture

Request: Whole leaf; fruit close-up; bud; flowers; multiple leaves

Assessment schedule

Assessment	Type	Items	Time
W04-PRACTICAL	substantial cumulative weekly practical	32	64 min
W04-NOMENCLATURE-2	cumulative nomenclature check	30	22 min
W04-SILVICS-2	cumulative silvics retrieval	16	40 min
W04-FIELD-CONTEXT	whole tree and site exercise	20	45 min

Five-session field workflow

Session A - Aspens and balsam poplar

organ comparisons across age and clonal form. Modalities: foliage, bark, dormant, whole tree context.

☐ Session A completed (enter X)

Session B - Populus and transition retrieval

unfamiliar isolated specimens and stand-context decisions. Modalities: foliage, bark, dormant, whole tree context.

☐ Session B completed (enter X)

Session C - Alder and mountain-ash comparisons

fruit foliage and growth-form contrasts. Modalities: foliage, reproductive, dormant, whole tree context.

☐ Session C completed (enter X)

Session D - Boreal-transition field laboratory

regional assemblage and single-organ stations. Modalities: foliage, bark, reproductive, dormant, whole tree context.

☐ Session D completed (enter X)

Session E - First substantial cumulative review

formal-practical preparation without image reuse. Modalities: foliage, bark, reproductive, dormant, whole tree context.

☐ Session E completed (enter X)

Evidence-to-decision protocol

Step	Field action
1. Inventory	Name only what is visible: organ, arrangement, attachment, surface, scale, maturity, damage, and context.
2. Generate	Name the leading identity and the nearest plausible alternative before seeking confirmation.
3. Separate	Cite the visible character that changes the decision; do not substitute habitat or color alone for structure.
4. Calibrate	Use species, genus, or insufficient evidence according to the photograph, not according to what you hope it shows.
5. Request	Ask for the single additional view with the highest expected diagnostic value.

Part I - Species instruction atlas

These profiles are labeled teaching material. Photographs come only from the teaching pool; no exact image in the formal assessment section appears here.

Quaking aspen - *Populus tremuloides*

New this week | Family: Salicaceae | Tier A | Priority 1 - core

Throughout the region, especially northern and boreal-transition landscapes

Extremely shade-intolerant clonal pioneer after fire or harvest



Bark

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Dormant

(c) nuthatchnicole, some rights reserved...
INAT_PHOTO_619021828 | cc-by



Foliage

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Evidence	Diagnostic interpretation
Foliage	Alternate, simple leaves on normal short shoots are usually nearly round to broadly ovate or rhombic, 2-8 cm long, with mostly 18-30 small rounded teeth per side. The long petiole is strongly flattened at right angles to the blade, allowing the leaf to quake in light wind. Vigorous root suckers and stump shoots may carry much larger, less circular, coarsely toothed leaves and must not be keyed as normal crown foliage.
Bark age series	Young stems have thin, smooth, pale greenish-white to cream bark, often with dark branch scars and horizontal lenticels. Pole bark becomes gray-green or yellowish, while old lower boles become gray to nearly black and deeply furrowed even though upper bark remains pale and smooth. Powdery surface bloom, wetness, lichens, cankers, and animal scarring can substantially alter the familiar white-bark appearance.
Dormant	Alternate slender brown to gray-red twigs end in narrow, pointed, usually glabrous buds with overlapping glossy scales; lateral buds are appressed. Leaf scars are alternate and small. Dormant twigs resemble other aspens, so the glabrous buds are most useful when directly contrasted with the visibly hairy buds of bigtooth aspen and the much larger, sticky, strongly aromatic buds of balsam poplar.
Fruit / cone / seed	Male and female catkins occur on separate clones and expand before the leaves. Female catkins develop many small capsules that split to release minute seeds embedded in cottony hairs in late spring or early summer. Catkins and cottony seed establish <i>Populus</i> well but usually need associated foliage, buds, or a known parent tree for a defensible species call.
Whole tree / context	A fast-growing, straight-stemmed small to medium canopy tree often occurring in even-aged clonal patches after fire, clearcutting, windthrow, or land abandonment. Pale upper boles, fine branches, leaves visibly trembling, and a dense cohort of connected stems support the call across northern hardwood, aspen-birch, and boreal-transition landscapes.
Variation / condition	Leaf size and tooth pattern differ sharply between crown short shoots and vigorous suckers. Natural hybrids with bigtooth aspen can show intermediate blade size and roughly 14-20 intermediate-height teeth per side. Hypoxylon canker, trunk decay, browsing, sapsucker holes, fire scars, old rough basal bark, and browse- or weevil-deformed crowns should be treated as condition evidence rather than different taxa.
Evidence limit	Several normal crown leaves showing many small teeth plus a strongly flattened petiole usually support species, especially with glabrous buds or a clonal whole-tree view. A single giant sucker leaf, white bark alone, distant yellow foliage, detached catkins, or cottony seed alone supports only <i>Populus</i> , an aspen group, or insufficient evidence. Intermediate tooth counts should trigger consideration of <i>Populus × smithii</i> rather than a forced parent-species call.

Confuser control

Bigtooth aspen - *Populus grandidentata*

New this week | Family: Salicaceae | Tier A | Priority 1 - core

Great Lakes through southern ON/QC, NY, New England, and NB, often warmer/drier than quaking aspen

Fast-growing disturbance tree regenerating strongly by root suckers



Bark

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Foliage

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Whole Tree Context

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Evidence	Diagnostic interpretation
Foliage	Alternate, simple blades on normal shoots are broadly ovate, usually 7-12 cm long, and bear roughly 5-12 conspicuously large, rounded to blunt teeth per side. Expanding leaves are densely white-woolly before becoming mostly glabrous; the petiole is strongly flattened at right angles to the blade. Use ordinary crown leaves, because sprouts and hybrid leaves can be substantially different.
Bark age series	Saplings have smooth thin olive-green to gray bark with lenticels and dark branch scars. Pole bark becomes dull gray or yellow-green; mature lower boles darken and develop thick irregular furrows and ridges while upper stems remain smoother. Old rough bark, wet surfaces, and pale lichens overlap both quaking aspen and balsam poplar.
Dormant	Alternate gray-brown twigs have a true terminal bud and appressed lateral buds. The several overlapping bud scales are distinctly hairy, a valuable winter separator from the normally glabrous buds of quaking aspen; flower buds are plumper. Bud hair must be seen at close range and should be paired with twig and leaf evidence when possible.
Fruit / cone / seed	Male and female catkins occur on separate trees and appear before leaf expansion. Female catkins mature into capsules whose minute seeds are carried by cottony hairs. Reproductive material strongly supports <i>Populus</i> but is rarely diagnostic between the native aspens without bud, emerging-leaf, or parent-tree evidence.
Whole tree / context	A rapidly growing medium tree with a straight bole, ascending branches, and a rounded or open crown, commonly forming root-suckering patches on disturbed sandy uplands, glacial outwash, ridges, and well-drained northern hardwood sites. It is regionally more associated with warm, dry uplands than quaking aspen, but the site ranges overlap.
Variation / condition	New leaves lose their striking white wool as they expand, and sucker leaves can be oversized or atypically toothed. Natural hybrids with quaking aspen have intermediate blade size and tooth number. Cankers, heart rot, animal browse, logging scars, fire, lichens, and old furrowed basal bark can obscure normal form.
Evidence limit	A strong species call uses multiple normal leaves with few very large teeth plus hairy winter buds or clearly woolly expanding foliage. One coarse-toothed sucker leaf, rough bark, a cottony catkin, or dry-upland context alone is insufficient. Intermediate leaves should be recorded as the quaking-bigtooth complex or possible <i>Populus</i> × <i>smithii</i> until additional shoots and buds can be examined.

Confuser control

Alternative	Visible separator	Resolving view
<i>Populus tremuloides</i>	Bigtooth aspen normally has 5-12 large teeth per leaf side, larger 7-12 cm blades, white-woolly emerging leaves, and hairy bud scales. Quaking aspen normally has 18-30 tiny teeth per side, smaller blades, and glabrous buds.	Several mature short-shoot leaves with a scale reference, a newly expanding leaf if available, and a sharply focused winter bud showing scale hairs.
<i>Populus balsamifera</i>	Balsam poplar has a drawn-out leaf tip, much finer teeth, only weakly flattened petioles, and large sticky aromatic buds. Bigtooth aspen has fewer coarse blunt teeth, strongly flattened petioles, and hairy rather than strongly viscid buds.	Terminal bud, petiole cross-section, full leaf margin, and lower leaf surface checked for orange resin streaks.
<i>Populus</i> × <i>smithii</i>	The natural quaking-aspen × bigtooth-aspen hybrid typically has intermediate leaves, mostly about 46-74 mm long with roughly 14-20 teeth per margin, and may backcross. No single intermediate character reliably assigns a parent species.	Representative foliage from several crown positions and stems in the patch, bud pubescence, emerging leaves, and, if operationally important, a voucher reviewed by a botanist.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Very shade intolerant; successful suckers and seedlings need nearly full overhead light and decline rapidly when overtopped.
Moisture / soils	Most characteristic on well-drained sandy or loamy uplands and glacial deposits, but best growth occurs where those soils retain adequate moisture and fertility; prolonged saturation and very droughty infertile sand limit productivity.
Geographic affinity	A northeastern and Great Lakes species centered farther south and on somewhat warmer sites than quaking aspen, with particular abundance in the Great Lakes-St. Lawrence region.
Associations	Occurs in aspen, aspen-birch, pine-hardwood, and northern-hardwood mixtures with quaking aspen, paper birch, red maple, white and red pine, northern red oak, sugar maple, and conifer or hardwood successors.
Regeneration	Tiny windborne seed can colonize exposed moist mineral soil, but root suckering following major overstory disturbance supplies most regeneration in established stands.
Vegetative reproduction	Established lateral roots produce abundant suckers after cutting, fire, or death of parent stems. The resulting patch is clonal, and its regeneration potential depends on root condition and stored reserves.
Seed / mast	Seed crops are frequent once reproductive age is reached, but individual seeds are short-lived and successful seedling years require the brief coincidence of dispersal with continuously moist bare seedbeds.
Establishment	Dense suckering requires full light, warm soil, and undamaged living roots; seedling establishment requires exposed moist mineral soil with little litter and minimal early competition.
Succession	Early-seral pioneer that rapidly reforests disturbed uplands, stabilizes soil, and shelters slower-growing pines, conifers, and northern hardwoods that commonly replace it in the absence of further disturbance.
Disturbance response	Clearcutting and stand-replacing fire usually stimulate strong root suckering if root systems survive. Partial canopy retention, repeated severe burning, root disturbance, rutting, or heavy browsing can reduce cohort density and vigor.
Longevity / growth	Fast growing during the first decades and generally short-lived, with many stems declining within roughly a century even though the root clone may persist longer.
Browse	Buds and young shoots are important winter food for moose, deer, hare, and grouse; repeated ungulate browsing can suppress sucker height and alter stem form.
Insects / disease	Forest tent caterpillar, leaf-feeding insects, borers, Hypoxylon and other cankers, root disease, and heart rots can cause mortality, breakage, and serious internal defect.
Management	Assess clone vigor and defect before retaining an aspen objective, protect roots from equipment damage, use cohort-scale rather than single-tree thinking, and recognize that dry-site presence does not guarantee acceptable growth.
Silvicultural systems	Complete even-aged removal or coppice-style treatment usually gives the best sucker response. Selection and shelterwood residuals suppress suckers and expose retained stems and roots to damage, so they do not efficiently perpetuate bigtooth aspen.

Personal field cue for Bigtooth aspen

learner-created retrieval hook

Balsam poplar - *Populus balsamifera*

New this week | Family: Salicaceae | Tier B | Priority 2 - regional

Northern Great Lakes, ON/QC, northern NY/New England, and NB on cold floodplains and wet sites

Boreal riparian/lowland pioneer with strongly aromatic buds



Bark

(c) Loïc Mathieu, some rights reserved...
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Dormant

(c) Derek, some rights reserved (CC BY)
INAT_PHOTO_253369899 | cc-by



Foliage

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Evidence	Diagnostic interpretation
Foliage	Alternate, simple blades are broadly ovate to lance-ovate with a long acute or short-acuminate tip, a broad-cuneate to rounded or shallowly heart-shaped base, and many small rounded teeth. The upper surface is glossy dark green; the lower surface is paler and often bears orange resin streaks. Petioles are round to only weakly compressed, not strongly flattened like the two aspens.
Bark age series	Young stems have smooth gray-green to brown bark with evident lenticels; pole bark remains relatively smooth and gray. Mature lower boles become darker gray-brown and develop broad furrows and ridges while the upper bole remains smoother. Old wet or lichen-covered bark is not reliably separable from other northern poplars.
Dormant	The large, sharply pointed terminal bud is conspicuously resinous, sticky, and strongly balsamic-smelling; lateral buds are appressed on round twigs. This is one of the strongest winter characters in the region, but cultivated balsam-poplar hybrids may share it. Do not taste or unnecessarily damage a specimen to test odor.
Fruit / cone / seed	Male and female catkins occur on separate trees before leaf-out. Female catkins bear two- or three-valved capsules that release many cottony seeds. Catkin sex and capsule details can aid hybrid work, but isolated cottony fruit does not safely separate balsam poplar from other <i>Populus</i> .
Whole tree / context	A fast-growing medium to large northern tree with an open crown, stout ascending branches, and often a straight bole on alluvial flats, river and lake shores, cold seepage areas, and moist boreal mixedwoods. It may form pure floodplain patches or occur with spruce, aspen, birch, willow, and alder.
Variation / condition	Orange leaf resin can be faint or absent, and sucker leaves may be unusually broad. Planted balm-of-Gilead material and the balsam-poplar × eastern-cottonwood hybrid <i>Populus</i> × <i>jackii</i> can combine sticky aromatic buds with broader triangular foliage. Cankers, rust, stem decay, flooding scars, beaver cutting, and wet bark can alter form and texture.
Evidence limit	Native balsam poplar is best supported by a large viscid aromatic bud together with long finely toothed leaves, weakly compressed petioles, round branchlets, and northern wet-site context. A sticky bud alone may indicate the balsam-poplar hybrid complex, while catkins, cottony seed, rough bark, or habitat alone support only <i>Populus</i> or insufficient evidence.

Confuser control

Alternative	Visible separator	Resolving view
<i>Populus tremuloides</i> and <i>Populus grandidentata</i>	Both aspens have strongly laterally flattened petioles and rounder blades; quaking aspen has many tiny teeth and bigtooth aspen has few very large teeth. Balsam poplar has a drawn-out leaf tip, finer teeth, a weakly compressed petiole, and a much larger sticky aromatic terminal bud.	Petiole cross-section, terminal bud with scale and resin visible, entire leaf margin, and lower leaf surface.
<i>Populus deltoides</i> and <i>Populus</i> × <i>jackii</i>	Eastern cottonwood has broad triangular leaves, petiole glands, and often angular young twigs. <i>Populus</i> × <i>jackii</i> can have triangular-ovate leaves, round twigs, and viscid balsam-like buds, so morphology may place it only in the hybrid complex.	Several long-shoot and short-shoot leaves, petiole apex checked for glands, twig cross-section, buds, capsules, and evidence of planting or nearby cultivated trees.
<i>Populus heterophylla</i>	Swamp cottonwood has broader leaves with a rounded or blunt apex, persistent hair along lower-surface veins, and a terminal bud that is scarcely viscid. Balsam poplar has an acute to short-acuminate tip, a mostly glabrous lower surface often streaked with orange resin, and a very viscid bud.	Leaf apex and base, magnified lower-surface veins, and an intact terminal winter bud.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Very shade intolerant; regeneration needs nearly full light and is rapidly overtopped beneath spruce, fir, cedar, or hardwood canopies.
Moisture / soils	Best developed on moist, nutrient-rich, aerated alluvial soils and fresh flood deposits; it also occupies moist uplands in the far north but performs poorly under stagnant anaerobic water or severe drought.
Geographic affinity	The northernmost native North American hardwood, strongly associated with boreal and cold continental climates; in the course region it is most characteristic of northern Great Lakes and Canadian lowlands and floodplains.
Associations	Occurs with white spruce, black spruce, balsam fir, quaking aspen, paper birch, black ash, tamarack, northern white-cedar, speckled alder, and riparian willows.
Regeneration	Abundant wind- and water-dispersed seed colonizes fresh moist silt, sand, and exposed mineral soil; root suckers and sprouts contribute strongly after cutting, burning, or stem injury on established sites.
Vegetative reproduction	Roots and stumps can produce vigorous sprouts, and suitable stem material roots more readily than aspen. Vegetative regeneration can dominate cutovers, although some under-canopy suckers are short-lived without release.
Seed / mast	Female trees produce large quantities of minute cottony seed at short intervals, but seed is short-lived; effective recruitment depends more on fresh moist substrate during dispersal than on a persistent seed bank.
Establishment	Successful seedlings require bare, continuously moist but aerated mineral or alluvial seedbeds in full light. Suckers require living roots, adequate reserves, and prompt crown release.
Succession	A primary pioneer on new river deposits and a secondary pioneer after fire or harvest; on many floodplains it dominates early decades and is gradually replaced by white spruce or other shade-tolerant associates.
Disturbance response	Flood deposition, erosion, fire, and overstory removal create suitable seedbeds or stimulate vegetative shoots. Severe root damage, long-term inundation, competition, and shaded partial cuts reduce successful renewal.
Longevity / growth	Fast growing and generally short-lived to moderately long-lived; many stems decline within roughly a century, although exceptional floodplain trees can approach two centuries.
Browse	Twigs and bark are heavily used by moose, deer, snowshoe hare, and beaver; repeated browsing and beaver cutting can strongly shape stand structure.
Insects / disease	Leaf rusts, Septoria and Cytospora cankers, borers, defoliators, root disease, and heart rots can reduce vigor and wood quality; wounds and flood stress increase opportunities for decay.
Management	Protect riparian soils and hydrology, avoid rutting shallow roots, retain wildlife and bank-stabilization functions, and distinguish native stands from planted hybrid poplars before drawing ecological conclusions.
Silvicultural systems	Even-aged overstory removal can release sprouts and suckers where balsam poplar is an objective. Natural regeneration on new alluvium depends on exposed seedbeds; partial shade favors replacement by spruce, fir, cedar, or other associates.

Personal field cue for Balsam poplar

learner-created retrieval hook

Speckled alder - *Alnus incana* subsp. *rugosa*

New this week | Family: Betulaceae | Tier B | Priority 2 - regional

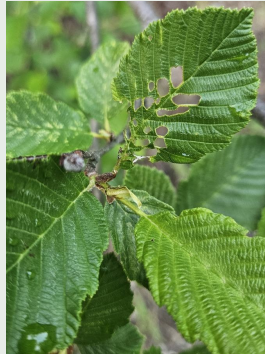
Wet thickets, stream margins, and peatland edges across the northern region

Nitrogen-fixing colonizer of disturbed wet ground and riparian succession



Dormant

(c) Bryce, some rights reserved (CC BY)
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Foliage

(c) willingarden, some rights reserved...
INAT_PHOTO_526024569 | cc-by



Mixed Evidence

(c) rboles, some rights reserved (CC BY)
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Evidence	Diagnostic interpretation
Foliage	Alternate, simple leaves are narrow-ovate to elliptic or broadly elliptic, commonly with a rounded to subcordate base, an acute tip, and coarse double serration. The lower surface often becomes glaucous, and strong straight side veins with conspicuous cross-veins give it a corrugated look. Several mature leaves are needed because shape and pubescence vary.
Bark age series	Young brown to gray stems are smooth and densely speckled with conspicuous horizontal or oval white lenticels, the source of the common name. Older multiple stems remain relatively thin-barked but become grayer and slightly scaly while retaining pale lenticels. Wet bark, lichens, and shade can hide or imitate the speckling.
Dormant	Alternate ovoid buds stand on obvious 2-4 mm stalks and are covered by two, sometimes three, roughly equal scurfy scales. Long pendulous staminate catkins and shorter pistillate catkins are preformed and exposed through winter, while old woody infructescences often persist. This combined bud-and-catkin architecture is much stronger than twig color alone.
Fruit / cone / seed	Staminate catkins elongate and shed pollen before leaf-out; smaller pistillate catkins mature into persistent woody cone-like strobiles. The small nutlets are merely thin-edged rather than broadly winged. Persistent cones prove <i>Alnus</i> , but close foliage, buds, bark, and geographic context are needed for subspecies-level identification.
Whole tree / context	Usually a many-stemmed shrub or small tree forming dense nearly continuous thickets on streambanks, lake margins, wet field edges, beaver meadows, swamps, and peatland margins. A low tangled crown, persistent alder cones, white-speckled stems, and saturated northern context together are characteristic.
Variation / condition	Leaf bases, degree of bloom, resin, and serration vary, and hybridization with smooth alder can produce subtle intermediates where ranges overlap. Coppicing, ice, beaver cutting, browse, floods, alder insects, lichens, and stem death create irregular thicket form without changing host identity.
Evidence limit	A defensible subspecies call combines abundant white lenticels, glaucous coarsely double-serrate leaves, and stalked two-scaled buds; persistent cones alone establish only <i>Alnus</i> . Sterile shoots lacking buds, blurred wetland panoramas, or plants intermediate with smooth alder should remain <i>Alnus incana</i> , <i>Alnus</i> sp., or insufficient evidence rather than being forced to subsp. <i>rugosa</i> .

Confuser control

Alternative	Visible separator	Resolving view
<i>Alnus serrulata</i>	Speckled alder normally has many pale bark lenticels, coarsely double-serrate leaves with a glaucous lower surface and conspicuous cross-veins, and little resin blotching. Smooth alder usually has fewer darker lenticels, more finely regular serration, no lower-surface bloom, and thin resin blotches; hybrids can be intermediate.	Multiple mature leaf undersides, close bark and twig lenticels, stalked winter buds, and the branchlet geometry supporting staminate and pistillate catkins.
<i>Alnus glutinosa</i>	European alder commonly becomes a larger single-stemmed tree and has broadly obovate to nearly round leaves with a rounded or notched apex. Speckled alder is usually a northern thicket shrub with elliptic leaves that have an acute tip and conspicuous pale stem lenticels.	Several mature leaves showing the apex, whole growth form, winter buds, fruiting stalks, and local planting or naturalization context.
<i>Betula populifolia</i> and wetland <i>Salix</i> species	Gray birch has papery chalky bark with dark chevrons and nonpersistent female catkins that disintegrate; willows have one bud scale and capsules rather than persistent woody cones. Speckled alder has stalked two-scaled buds and cone-like infructescences.	A close winter bud, attached persistent reproductive structure, mature bark, and complete leaf attachment.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Generally shade intolerant to intermediate; dense thickets persist where crowns receive open or edge light but thin beneath a closed forest canopy.
Moisture / soils	Characteristic of saturated or seasonally flooded mineral and organic soils along streams, shorelines, swamp margins, wet fields, and peatland edges; it tolerates poor drainage better than most course taxa.
Geographic affinity	A cold-temperate and boreal North American subspecies widespread across the northern Great Lakes, southern Canadian, northeastern, and Acadian parts of the course region.
Associations	Occurs with willows, balsam poplar, red maple, black ash, tamarack, northern white-cedar, black spruce, sedges, and early riparian or beaver-meadow vegetation.
Regeneration	Small nutlets disperse by wind and water from persistent cones and establish on exposed wet mineral or organic substrates, especially after flooding, drawdown, or soil disturbance.
Vegetative reproduction	Rhizomes, root-crown shoots, and basal shoots allow patches to expand and recover after cutting, browsing, ice, or top-kill; individual cut stems die, but the connected plant can send up replacement stems.
Seed / mast	Cones mature in autumn and release numerous small nutlets through fall and winter. Crop size varies, and establishment depends on wet exposed substrate rather than a dependable long-lived seed bank.
Establishment	Moist exposed soil, high water availability, and adequate light favor germination and thicket expansion. Seedlings and new shoots are less competitive under dense shade or on dry sites.
Succession	Early-seral wetland and riparian colonizer whose Frankia root nodules fix atmospheric nitrogen, enrich local soils, stabilize banks, and sometimes precede willow, poplar, spruce, cedar, ash, or other later vegetation.
Disturbance response	Flooding, drawdown, beaver activity, cutting, and other wet-site disturbance can create seedbeds or stimulate vegetative shoots. Fire usually top-kills thin stems, but surviving belowground organs may resprout where organic soils and roots are not consumed.
Longevity / growth	Individual stems are relatively short-lived and modest in diameter, while a multi-stemmed rhizomatous patch can persist through repeated stem replacement.
Browse	Twigs and foliage are browsed by moose, deer, hare, and beaver; heavy use can suppress height, create dense low shoots, or repeatedly remove older stems.
Insects / disease	Alder flea beetles, sawflies, leaf miners, defoliating caterpillars, cankers, and stem or root rots occur; damage is often conspicuous locally but established clones can replace affected stems.
Management	Retain where bank stability, wetland cover, nitrogen inputs, and wildlife browse are objectives; where alder blocks desired conifer or hardwood regeneration, expect repeated treatment because cutting one stem does not eliminate the clone and observe wetland rules.
Silvicultural systems	Conventional overstory systems apply poorly to this shrub. Opening or wet-ground disturbance can increase thickets, whereas sustained canopy shade can reduce them; control or retention should be planned at patch scale with hydrology and repeated vegetative response in mind.

Personal field cue for Speckled alder

learner-created retrieval hook

American mountain-ash - *Sorbus americana*

New this week | Family: Rosaceae | Tier B | Priority 2 - regional

Cool uplands from MN/WI/MI through ON/QC, NY/New England, and NB

Bird-dispersed boreal/montane small tree of gaps and edges



Dormant

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Foliage

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Mixed Evidence

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Evidence	Diagnostic interpretation
Foliage	Leaves are alternate and pinnately compound, generally 10-22 cm long, with a terminal leaflet and numerous narrowly lanceolate to oblong sharply serrate leaflets. Typical leaflets are about three to five times as long as wide and mostly glabrous, including the lower surface. Always show the attachment of the entire compound leaf because a detached leaflet can resemble a simple-leaved tree.
Bark age series	Young trunks are thin, smooth, gray to gray-brown, with horizontal lenticels; older small boles become somewhat scaly or shallowly fissured but rarely develop massive ridges. Browse scars, fire blight, lichens, and multiple stems can make bark especially unreliable for separating the native mountain-ashes.
Dormant	Alternate twigs carry a conspicuous ovoid to pointed terminal bud and smaller lateral buds with overlapping scales. Twigs, bud scales, and leaflet undersides are glabrous or only very sparsely hairy; bud-scale margins are not or only sparsely ciliate. Leaf scars are alternate. Bud size and hair require a macro view and direct comparison.
Fruit / cone / seed	Many small five-petaled white flowers form a broad flat-topped corymb in late spring or early summer. Orange-red to red pomes about 4-7 mm in diameter ripen in late summer and can persist into winter; each retains a small calyx crown. Fruit size must be measured because perspective and crop condition are misleading.
Whole tree / context	A slow-growing shrub or small tree, usually with a short trunk, slender spreading branches, and a narrow open rounded crown in cool humid openings, spruce-fir and northern-hardwood edges, rocky ridges, swamps, and montane sites. Bird perches, disturbed edges, and semiopen stands are more characteristic than deep closed shade.
Variation / condition	Leaflet width, tip, glaucescence, fruit color, and crown form vary with exposure and shoot vigor. Moose and deer can maintain old plants as low distorted browse, while fire blight, rust, borers, storm breakage, and fruit removal by birds can erase familiar characters. Ornamental European rowan can escape near settlements.
Evidence limit	Species-level identification requires an alternate compound leaf plus the narrow-leaflet, low-hair suite and preferably measured fruit, petals, or bud-scale margins. A distant red-fruited crown, smooth gray bark, one detached leaflet, or an unmeasured fruit cluster supports only <i>Sorbus</i> or insufficient evidence. Sterile plants should often remain the <i>S. americana</i> - <i>S. decora</i> complex.

Confuser control

Alternative	Visible separator	Resolving view
<i>Sorbus decora</i>	American mountain-ash typically has leaflets three to five times as long as wide, bud-scale margins not or sparsely ciliate, 3-4 mm petals, and 4-7 mm fruit. Showy mountain-ash tends to have leaflets only two to three-and-a-half times as long as wide, ciliate bud-scale margins, 4-5 mm petals, and 7-10 mm fruit.	Several complete leaves, macro photographs of terminal-bud scale margins, and flowers or ripe fruit photographed with a ruler.
<i>Sorbus aucuparia</i>	European rowan usually has evident hair on young branchlets, inflorescences, bud scales, and lower leaflet surfaces. American mountain-ash is glabrous or very sparsely hairy in those parts; planted and escaped context increases the probability of European rowan.	Sharp close-ups of young branchlets, flower-cluster axes, bud scales, and both leaflet surfaces, plus evidence of cultivation or naturalized cohorts.
<i>Fraxinus</i> species	True ashes have opposite compound leaves and opposite twigs, one-seeded samaras, and no red pomes. Mountain-ash has alternate leaves and buds plus flat clusters of calyx-crowned pomes.	Several consecutive twig nodes showing arrangement and any attached fruit.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Shade intolerant overall; it can persist beneath a semiopen canopy but abundance, flowering, and recruitment decline in deep closed shade.
Moisture / soils	Most successful on cool moist acidic soils with adequate aeration, but occurs from swamp borders to well-drained rocky slopes and can survive in stunted form on drier sites.
Geographic affinity	A northeastern cold-temperate, boreal, and montane species extending from the Great Lakes and southern Canada through New England and the higher Alleghenies and Appalachians.
Associations	Occurs in spruce-fir, aspen-birch, northern hardwood, and pine mixtures, often with balsam fir, red or white spruce, paper birch, mountain maple, serviceberry, and other gap or edge shrubs.
Regeneration	Birds and small mammals disperse the pomes and seed; seeds require an extended cold stratification period, and seedlings recruit most readily in moist semiopen microsites.
Vegetative reproduction	Stumps can sprout after top-kill, allowing established individuals to survive some logging, browse, or fire damage, but off-site bird-dispersed seed is also important for colonization.
Seed / mast	Fruit crops are recurrent but variable, ripen in late summer, and may remain available into winter until removed by birds and mammals. The course should not assume a fixed mast interval.
Establishment	Cold stratification, a moist seedbed, and enough gap or edge light for early growth are more important than mineral-soil exposure alone; seedlings are vulnerable to drying and browsing.
Succession	Facultative early-seral or gap species that becomes more abundant after opening yet can persist at low density in old spruce-fir and northern-hardwood stands.
Disturbance response	Logging and canopy mortality commonly increase surviving shoots and light; thin bark makes stems vulnerable to fire, but root-crown or stump sprouts and later bird-dispersed seedlings can reoccupy disturbed sites.
Longevity / growth	Slow growing, short-lived, and usually confined to the understory or small-tree layer rather than competing for upper-canopy dominance.
Browse	Highly preferred by moose and white-tailed deer; repeated browsing can hold stems below recruitment height for decades and locally eliminate mature fruiting trees despite persistent small sprouts.
Insects / disease	Fire blight, rusts, leaf spots, apple scab, sawflies, borers, and defoliators can damage foliage, shoots, and fruit; stressed small trees also suffer stem decay and breakage.
Management	Retain as a wildlife-fruit and browse component where compatible with objectives, protect a portion of stems from excessive ungulate pressure, and do not infer conifer-site quality from its presence alone.
Silvicultural systems	Group openings, edge creation, and partial canopy release can increase vigor and fruiting. Heavy conifer release treatments may remove it as competition, so balance crop-tree goals against its disproportionate browse and soft-mast value.

Personal field cue for American mountain-ash

learner-created retrieval hook

Showy mountain-ash - *Sorbus decora*

New this week | Family: Rosaceae | Tier B | Priority 2 - regional

Boreal/Acadian areas around Lake Superior, northern ON/QC, northern NY/New England, and NB

Boreal small tree of cool moist sites with wildlife fruit



Foliage

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Mixed Evidence

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Foliage

(c) Clifton Ladd, C.W.B., some rights...
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Evidence	Diagnostic interpretation
Foliage	Alternate pinnately compound leaves commonly carry 13-17 blue-green leaflets. Compared with American mountain-ash, the leaflets are generally shorter and broader, about two to three-and-a-half times as long as wide, with serrate margins and acute to sometimes acuminate tips. A single plant may carry both blunt and pointed leaflet tips, so tip shape is not independently diagnostic.
Bark age series	Young gray-brown bark is smooth with lenticels; older small boles become scaly or shallowly fissured but remain thin. It closely overlaps American mountain-ash and European rowan at every age and should be used only with foliage, buds, or measured reproductive material.
Dormant	Alternate twigs end in a conspicuous ovoid to pointed scaly bud. The margins of the winter bud scales are distinctly ciliate, unlike the nonciliate or very sparsely ciliate scales typical of American mountain-ash. Hair and cilia require a sharp macro view; distant bud color or outline is not enough.
Fruit / cone / seed	Broad corymbs of white five-petaled flowers appear in late spring, about 7-10 days earlier than American mountain-ash where they grow together. Petals are about 4-5 mm long, and the showy red pomes are usually about 7-10 mm in diameter and often persist into winter. Scale is essential when using fruit size.
Whole tree / context	A shrub or small tree to roughly 10 m, with a rounded crown in boreal and Acadian forest openings, cool moist woods, swamps, rocky ridges, shores, and subalpine sites. It is strongly plausible around Lake Superior, northern Ontario and Quebec, northern New England, and New Brunswick, but context cannot separate it from American mountain-ash.
Variation / condition	Leaflet apex and lower-surface bloom vary within individuals. Plants sometimes separated as <i>Sorbus groenlandica</i> cannot be reliably distinguished morphologically and are included here for the course. Browse, cold exposure, fire blight, rust, stem injury, ornamental planting, and heavy fruit removal can alter crown, foliage, and reproductive display.
Evidence limit	A defensible species call uses multiple broad leaflets plus ciliate bud-scale margins or measured flowers and fruit; northern habitat and blue-green foliage are corroborative only. Bark, one detached leaflet, fruit color, or a distant cluster cannot reliably separate <i>S. decora</i> from <i>S. americana</i> or planted <i>S. aucuparia</i> , and sterile material often warrants <i>Sorbus</i> sp. or the native mountain-ash complex.

Confuser control

Alternative	Visible separator	Resolving view
<i>Sorbus americana</i>	Showy mountain-ash generally has leaflets two to three-and-a-half times as long as wide, ciliate winter-bud scale margins, 4-5 mm petals, and 7-10 mm fruit. American mountain-ash has narrower leaflets three to five times as long as wide, little or no bud-scale ciliation, 3-4 mm petals, and 4-7 mm fruit.	Several complete leaves, a macro view of terminal-bud scale margins, and flowers or mature fruit photographed beside a ruler.
<i>Sorbus aucuparia</i>	European rowan typically shows conspicuous hair on young branchlets, inflorescence axes, bud scales, and lower leaflet surfaces and is frequent in cultivated or escaped settings. Showy mountain-ash may have ciliate bud margins but is not densely hairy across that entire character suite.	Young twig and inflorescence axes, both leaflet surfaces, terminal bud, and a whole-site view showing planting or naturalized context.
<i>Fraxinus</i> and <i>Amelanchier</i> species	Ashes have opposite compound leaves and samaras, while serviceberries have alternate simple leaves and dark calyx-crowned pomes. Showy mountain-ash combines alternate pinnate leaves with dense clusters of red pomes.	Several consecutive nodes, one complete leaf from petiole base to terminal leaflet, and an attached fruit cluster.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Tolerates partial shade but not a deeply closed canopy; crown growth, flowering, and fruiting are best in openings and along forest edges.
Moisture / soils	Favored by cool moist conditions and occurs on a variety of acidic to neutral substrates, including rocky or relatively poor soils; heat, drought, and warm lowland exposure are important limitations.
Geographic affinity	A distinctly northern boreal, Acadian, and subalpine mountain-ash centered in Canada and the northern Great Lakes and New England portions of the course region.
Associations	Occurs in openings and edges of spruce-fir, boreal mixedwood, aspen-birch, and cool northern-hardwood forests, as well as swamps, shores, ridges, and subalpine vegetation.
Regeneration	Birds and small mammals disperse the fleshy pomes. As in regional Sorbus, seed dormancy is overcome by seasonal stratification, and recruitment is most likely in moist semiopen microsites.
Vegetative reproduction	Basal or stump sprouting may allow damaged mountain-ash stems to recover, but species-specific evidence for the frequency and stand importance of vegetative reproduction in S. decora is limited; do not assume an aspen-like clonal strategy.
Seed / mast	Conspicuous crops ripen in late summer and often remain into winter, but production varies with crown light, weather, and disease. No fixed course mast interval should be inferred from one heavy fruit year.
Establishment	Cold conditioning, a reliably moist seedbed, and partial to high light in a cool setting favor recruitment; warm droughty sites and deep shade limit establishment and subsequent growth.
Succession	A gap, edge, and disturbed-site small tree that can persist as a minor component of mature boreal or Acadian forest but does not become a dominant upper-canopy species.
Disturbance response	Canopy gaps, shores, and moderate soil disturbance may improve light and seedling opportunities. Thin bark and small stature make stems vulnerable to fire and mechanical injury; the relative roles of resprouting and seed after disturbance are not well quantified for this species.
Longevity / growth	Slow to moderate growing, modest in stature, and generally short-lived compared with canopy spruce, hemlock, or northern hardwoods; wood and branches are relatively weak.
Browse	Twigs and foliage are used by deer and moose, and repeated browsing can deform or suppress young stems; persistent fruits are important winter food for birds and small mammals.
Insects / disease	Fire blight, leaf spots, rusts, sawflies, borers, and other rosaceous-tree pests can damage foliage and shoots, especially on stressed trees, though it is often considered more disease-resistant in cultivation than European rowan.
Management	Retain for northern forest diversity and durable wildlife fruit where compatible with objectives, protect selected stems from chronic browse, and verify identity before using it as an indicator of boreal site conditions.
Silvicultural systems	Small group openings, selection edges, and retention along cool moist ecotones can maintain crown light and fruiting. Full exposure on hot dry sites or complete suppression under dense residual canopy is unfavorable; species-specific yield prescriptions are not established.

Personal field cue for Showy mountain-ash

learner-created retrieval hook

Part II - Cumulative rapid-reference atlas

Use this as a field briefing, not as a substitute for retrieval. Earlier taxa are compressed here so the packet remains self-contained while preserving space for unfamiliar assessment images.

Taxon	High-value visible evidence	Silvics anchor
Sugar maple <i>Acer saccharum</i>	Foliage: Opposite, simple blades are palmately veined and usually five-lobed, with pointed lobes, broad rounded U-shaped... Bark: Seedlings and saplings have smooth light-gray bark; pole trees develop long shallow furrows; mature and old... Dormant: Opposite brown twigs carry a narrow, sharply pointed terminal bud with many tightly overlapping...	Tolerance: Very shade tolerant; seedlings and saplings can persist for years as advance... Site: Best on cool, fertile, moist but well-drained loams with adequate calcium and magnesium; performs...
Red maple <i>Acer rubrum</i>	Foliage: Opposite, simple blades usually have three to five lobes, angular or V-shaped sinuses, and regular serration... Bark: Young stems are smooth and light gray; intermediate bark develops narrow vertical cracks and small scaly plates;... Dormant: Opposite reddish twigs bear blunt-ovoid red buds with several pairs of scales; flower buds are...	Tolerance: Moderately shade tolerant when young and able to persist beneath partial canopy... Site: Occupies an exceptionally broad range from saturated swamps and floodplains to dry ridges; local seed...
American beech <i>Fagus grandifolia</i>	Foliage: Alternate, simple leaves are elliptic to ovate with a short petiole, straight parallel lateral veins, and one... Bark: Saplings and healthy mature trees have thin, smooth, pale bluish-gray bark, often compared with elephant hide... Dormant: Alternate twigs carry exceptionally long, slender, sharply pointed cigar-shaped buds with many...	Tolerance: Exceptionally shade tolerant; seedlings, saplings, and root suckers can persist... Site: Most competitive on moist, well-drained, often fertile loams and slopes; absent from persistently...
Yellow birch <i>Betula alleghaniensis</i>	Foliage: Alternate ovate leaves have a rounded to weakly heart-shaped base, many straight lateral veins, and sharply... Bark: Seedlings are gray-brown and unremarkable; intermediate stems develop bronze to yellow-gold bark with horizontal... Dormant: Slender alternate twigs bear pointed reddish-brown buds, small spur shoots, and no true terminal...	Tolerance: Intermediate in shade tolerance: more tolerant than paper birch and aspen, but... Site: Best on cool, moist, well-drained, fertile loams and coves; common on north and east slopes, seepage...
Paper birch <i>Betula papyrifera</i>	Foliage: Alternate ovate to triangular leaves have a rounded or wedge-shaped base, pointed tip, and doubly serrate... Bark: Young stems are reddish-brown with pale horizontal lenticels and do not look white; pole and mature bark becomes... Dormant: Slender reddish-brown alternate twigs have pointed appressed buds, conspicuous resin glands or...	Tolerance: Very shade intolerant; seedlings require abundant light and are rapidly overtopped... Site: Occupies cool sites across a broad moisture range, from well-drained glacial tills and sandy soils to...
Eastern hemlock <i>Tsuga canadensis</i>	Foliage: Short, flat, mostly blunt single needles vary in length along delicate twigs, show two pale stomatal bands... Bark: Young bark is smooth gray-brown; intermediate stems develop narrow scaly ridges; mature bark becomes dark gray... Dormant: Fine flexible yellow-brown twigs carry minute rounded buds, tiny petiolate needles, and often a...	Tolerance: Extremely shade tolerant, among the most tolerant North American trees; seedlings... Site: Favors cool, humid, sheltered sites with reliable moisture and generally well-drained acidic to...
American basswood <i>Tilia americana</i>	Foliage: Alternate, broadly ovate to heart-shaped leaves have an abruptly pointed tip, serrate margin, palmate basal... Bark: Young bark is smooth gray to gray-brown; intermediate bark develops narrow vertical fissures; old boles are dark... Dormant: Alternate zigzag twigs carry plump red to reddish-brown ovoid buds with only two or three...	Tolerance: Shade tolerant, though generally less persistent in deep shade than beech or sugar... Site: Best on deep, fertile, moist, well-drained loams with high base status, especially calcium-rich tills...
Black cherry <i>Prunus serotina</i>	Foliage: Alternate, simple, elliptic leaves have fine rounded or incurved teeth, one or more small glands near the top of... Bark: Young stems are smooth reddish-brown to dark gray with prominent horizontal lenticels; intermediate bark breaks... Dormant: Slender alternate reddish-brown twigs carry small ovoid scaled buds and semicircular leaf scars;...	Tolerance: Shade intolerant; seedlings may survive briefly under partial canopy, but... Site: Best on deep, fertile, moist but well-drained loams, especially Allegheny uplands and protected...
Eastern hophornbeam <i>Ostrya virginiana</i>	Foliage: Alternate ovate leaves have many straight lateral veins, a sharply pointed tip, and a doubly serrate margin; the... Bark: Young stems are smooth gray-brown with lenticels; older small trunks break into narrow, loose, vertical gray... Dormant: Slender alternate twigs bear short pointed ovoid buds with many visible scales; several...	Tolerance: Shade tolerant and able to persist as a slow-growing understory tree beneath... Site: Common on dry-mesic to mesic, well-drained, often rocky or calcareous soils; avoids prolonged flooding...
Black maple <i>Acer nigrum</i>	Foliage: Opposite blades are usually three-lobed, broad and somewhat drooping, with a deeply heart-shaped base, few... Bark: Young bark is gray and fairly smooth; older boles become dark gray-brown with irregular furrows and firm plates... Dormant: Opposite brown twigs carry a pointed terminal bud and smaller paired laterals. Bud, twig, and...	Tolerance: Very shade tolerant when young, broadly like sugar maple; suppressed advance... Site: Best on fertile, moist, well-drained loams and alluvium, often over calcareous or otherwise base-rich...
Striped maple <i>Acer pensylvanicum</i>	Foliage: Opposite, large, thin blades have three shallow forward-pointing lobes, a broad heart-shaped base, and fine... Bark: Saplings have smooth green bark marked by conspicuous vertical white stripes; older stems turn greenish brown to... Dormant: Opposite green to green-brown twigs and young stems retain longitudinal pale stripes; paired...	Tolerance: Very shade tolerant; survives decades suppressed but grows best under moderate... Site: Favors cool, moist, well-drained sandy loams, commonly acidic, and avoids both stagnant wetness and...
Mountain maple <i>Acer spicatum</i>	Foliage: Opposite blades are usually three-lobed, sometimes weakly five-lobed, with sharply pointed lobes and coarse... Bark: Young stems are reddish to gray-brown without white longitudinal stripes; older small trunks remain thin-barked... Dormant: Slender opposite twigs carry small pointed paired buds and a terminal bud; stems lack striped...	Tolerance: Moderately shade tolerant and able to persist beneath canopy, but density and... Site: Favors cool, moist, well-drained upland soils, rocky slopes, ravines, and streamside benches; drought...
Norway maple <i>Acer platanoides</i>	Foliage: Opposite, broad blades usually have five sharply pointed lobes and sparse large teeth. A freshly broken petiole... Bark: Young bark is smooth gray; mature bark develops regular narrow vertical ridges and shallow interlacing furrows... Dormant: Opposite stout twigs bear large plump terminal buds and paired lateral buds, commonly reddish...	Tolerance: Seedlings are shade tolerant to very shade tolerant and form persistent seedling... Site: Broadly site tolerant, performing especially well on cool moist rich soils while also tolerating urban...
Heart-leaved paper birch <i>Betula cordifolia</i>	Foliage: Alternate ovate leaves have a distinctly heart-shaped base, a pointed but not long-drawn tip, double teeth, and... Bark: Young stems are reddish to brown; mature bark becomes pink-white, brown-white, or red-brown-white and exfoliates... Dormant: Alternate twigs lack a true terminal bud and carry small ovoid imbricate buds; short shoots...	Tolerance: Generally shade intolerant to intermediate; seedlings and crowns perform best in... Site: Favors cool, moist, well-drained, often rocky or talus-derived upland soils and is poorly suited to...
Gray birch <i>Betula populifolia</i>	Foliage: Alternate glossy blades are triangular to rhombic-triangular with a long narrow tip, coarse double teeth, and... Bark: Young bark is brown to reddish; saplings and small trees become chalky gray-white with dark triangular chevrons... Dormant: Slender alternate twigs lack a true terminal bud, show lenticels and small resin glands, and...	Tolerance: Very shade intolerant; recruitment and persistence require open conditions. Site: Grows from moist well-drained margins to dry infertile sands, gravels, rocky slopes, and acidic...

Taxon	High-value visible evidence	Silvics anchor
Sweet birch <i>Betula lenta</i>	Foliage: Alternate ovate to narrow-ovate leaves are finely and regularly double-serrate, usually with at least six teeth... Bark: Young bark is smooth reddish brown with horizontal lenticels and can resemble cherry; mature bark becomes dark... Dormant: Slender reddish-brown twigs lack a true terminal bud and carry small pointed imbricate buds; a...	Tolerance: Classed as shade intolerant overall, although seedlings benefit from side shade or... Site: Best on moist well-drained soils and protected north- or east-facing slopes; tolerates rocky shallow...
Pin cherry <i>Prunus pensylvanica</i>	Foliage: Alternate thin leaves are narrow-ovate to lanceolate, long-pointed, finely serrate, and largely glabrous; one or... Bark: Young bark is shiny reddish brown with strong horizontal lenticels; older thin bark becomes gray-brown and may... Dormant: Slender reddish-brown twigs show horizontal lenticels and small sharp imbricate buds, including a...	Tolerance: Very shade intolerant; establishment and survival are concentrated in large openings. Site: Occupies coarse to medium soils from rocky ledges and sands to moist loams, but is generally absent...
Chokecherry <i>Prunus virginiana</i>	Foliage: Alternate oblong to obovate leaves have sharply pointed teeth, mostly 8-11 pairs of lateral veins, and a mostly... Bark: Young stems are reddish to gray-brown with horizontal lenticels; older small trunks remain relatively thin and... Dormant: Slender brown to reddish twigs have obvious lenticels and sharp imbricate buds. The winter twig...	Tolerance: Shade tolerant enough to persist under forest canopy, but reaches greatest density... Site: Broadly adaptable from mesic forest edges and riparian terraces to dry-mesic thickets, provided...
American hornbeam <i>Carpinus caroliniana</i>	Foliage: Alternate elliptic to ovate leaves have a symmetrical rounded or shallowly cordate base, strong straight... Bark: Young and mature bark remains thin, smooth, blue-gray to slate gray, while the trunk develops distinctive sinewy... Dormant: Slender alternate twigs lack a true terminal bud; narrow ovoid buds have many overlapping scales...	Tolerance: Very shade tolerant as a seedling and sapling; tolerance declines with age, so... Site: Best on rich loams with high water-holding capacity, from somewhat poorly drained to well drained...
Pagoda dogwood <i>Cornus alternifolia</i>	Foliage: Leaves are alternate, though crowded near shoot tips, with entire to slightly wavy margins and several strong... Bark: Young stems are smooth reddish to green-brown; mature bark becomes gray-brown with shallow ridges or small... Dormant: Leaves and buds are alternate, often clustered near upturned branch tips; slender twigs are...	Tolerance: Shade tolerant and able to dominate locally in mature-forest understories, while... Site: Favors cool, rich, moist, well-drained forest soils and edges; drought and heat stress increase decline.
Smooth serviceberry <i>Amelanchier laevis</i>	Foliage: Alternate ovate to elliptic leaves are sharply toothed; at flowering they are already partly expanded and... Bark: Young stems and twigs are reddish; older bark remains thin and smooth gray with pale vertical stripes, later... Dormant: Slender alternate reddish-brown twigs have narrow pointed imbricate buds, a terminal bud,...	Tolerance: Tolerates understory shade but flowers, fruits, and develops best with partial sun... Site: Prefers moist, moderately to well-drained acidic to neutral soils and tolerates rocky forest sites if...
Balsam fir <i>Abies balsamea</i>	Foliage: Needles are single, soft, flattened, aromatic when crushed, and usually arranged in two lateral ranks on lower... Bark: Sapling bark is thin, smooth, gray, and commonly dotted with conspicuous raised resin blisters. Intermediate... Dormant: Slender gray-brown twigs are relatively smooth after needles fall, with round flush leaf scars...	Tolerance: Very shade tolerant; seedlings can persist for decades as advance regeneration and... Site: Occupies moist cool sites from poorly drained organic and fine-textured lowlands to well-drained...
White spruce <i>Picea glauca</i>	Foliage: Needles are single, stiff, sharp, four-angled to somewhat flattened, and roll between the fingers; each stands... Bark: Young bark is thin, smoothish, gray to brown, becoming scaly early. Intermediate and mature bark forms thin... Dormant: Current branchlets are light brown to pink-brown, usually hairless, and coated with a thin...	Tolerance: Intermediate in shade tolerance; more tolerant than jack or red pine and the... Site: Grows across a wide moisture range but performs best on moist, well-drained, relatively fertile loams...
Black spruce <i>Picea mariana</i>	Foliage: Short single needles are four-angled, stiff, dull gray- to blue-green, glaucous, and borne around the twig on... Bark: Saplings have thin gray-brown bark with fine scales; intermediate and mature stems become dark gray to reddish... Dormant: Young yellow-brown branchlets lack bloom and are minutely hairy; at least some hairs are gland...	Tolerance: Shade tolerant, especially as a seedling, but generally less tolerant than balsam... Site: Dominant on cold acidic peatlands, poorly drained organic soils, and nutrient-poor mineral sites,...
Red spruce <i>Picea rubens</i>	Foliage: Single sharp four-angled needles stand around the twig on woody pegs and roll between the fingers. They are... Bark: Young bark is smoothish gray-brown to reddish brown and soon develops thin flaky scales. Mature bark becomes... Dormant: Yellow-brown current branchlets are minutely hairy without a glaucous bloom; the hairs are...	Tolerance: Very shade tolerant, broadly comparable to balsam fir and more tolerant than white... Site: Best in cool humid climates on moist acidic, commonly shallow or stony soils with thick organic...
Norway spruce <i>Picea abies</i>	Foliage: Needles are single, stiff, sharp, four-angled, dark green, and borne on woody pegs around the shoot. They... Bark: Young bark is smoothish gray-brown to reddish brown, becoming finely scaly. Older trunks develop gray to brown... Dormant: Reddish-brown branchlets carry prominent woody needle pegs and pointed reddish-brown, usually...	Tolerance: Moderately shade tolerant as a young tree, but plantation growth and crown... Site: Prefers cool, moist, well-drained, commonly acidic loams or sandy loams; prolonged saturation, heat,...
Eastern white pine <i>Pinus strobus</i>	Foliage: Soft, slender, flexible blue-green needles occur in fascicles of five, one for each letter in 'white'; they are... Bark: Seedlings and young saplings have smooth thin gray-green bark. Intermediate stems become gray with shallow... Dormant: Twigs are slender and gray-brown, with a conspicuous terminal cluster of long narrow pointed...	Tolerance: Intermediate in shade tolerance; seedlings tolerate substantial shelter, but... Site: Occurs on many soils and is competitive on sandy, rocky, or glacial deposits; best growth is on moist,...
Red pine <i>Pinus resinosa</i>	Foliage: Needles occur in fascicles of two, are dark green, relatively straight, and usually 9-16 cm long. They are... Bark: Young stems have thin gray to reddish scaly bark. Intermediate and mature boles develop conspicuous orange-red... Dormant: Stout orange-brown branchlets end in clustered ovoid red-brown buds, usually resinous. Persistent...	Tolerance: Shade intolerant; seedlings and saplings need nearly full light, and crowns recede... Site: Best known on well-drained sands, gravels, and rocky ridges, though highest productivity occurs on...
Jack pine <i>Pinus banksiana</i>	Foliage: Needles occur in fascicles of two, are short, stiff, yellow- to gray-green, and usually 2-5 cm long. Each pair... Bark: Sapling bark is thin, dull gray to reddish brown, and finely scaly. Mature stems remain relatively thin-barked... Dormant: Slender gray-brown twigs bear small ovoid resinous buds and persistent paired fascicle sheaths...	Tolerance: Very shade intolerant, among the least tolerant northern conifers; seedlings... Site: Most competitive on dry nutrient-poor sands, dunes, outwash, and barrens, but also occurs on rocky...
Scots pine <i>Pinus sylvestris</i>	Foliage: Needles occur in fascicles of two, are stiff, noticeably twisted, often blue- or gray-green, and usually 3-7 cm... Bark: Young branches and upper trunks develop thin flaky cinnamon- to bright orange-brown bark, a major field clue... Dormant: Gray- to orange-brown twigs bear clustered ovoid pointed gray-brown to reddish buds that may...	Tolerance: Shade intolerant; regeneration and good crown growth require open sun. Site: Adaptable to nutrient-poor, acidic, sandy, and seasonally dry soils if drainage is good; it performs...
Tamarack <i>Larix laricina</i>	Foliage: Soft, slender, blue- to bright-green needles occur singly on elongating current shoots and in dense tufts of... Bark: Young bark is smoothish gray to reddish brown and soon becomes finely scaly. Mature bark sheds as small thin... Dormant: Winter crowns are completely leafless but retain numerous rounded buds on short dark knobby spur...	Tolerance: Very shade intolerant; seedlings may endure light shade briefly but must reach the... Site: Characteristic of bogs, fens, and cold swamps on organic soils, yet best growth often occurs on moist,...
European larch <i>Larix decidua</i>	Foliage: Soft bright-green deciduous needles occur singly on long shoots and in dense tufts on knobby short shoots... Bark: Young branch bark is smooth gray to yellow-brown. Mature trunks develop thick reddish- to gray-brown, deeply... Dormant: Leafless winter twigs carry many rounded buds on stout knobby spur shoots; lateral branchlets...	Tolerance: Very shade intolerant; best stem and crown development require full sun, though... Site: Prefers cool sites with moist, well-drained, moderately fertile loams or gravelly soils; waterlogging...

Taxon	High-value visible evidence	Silvics anchor
Northern white-cedar <i>Thuja occidentalis</i>	Foliage: Tiny opposite scale leaves overlap in four ranks and completely hide the twig, forming strongly flattened fan-... Bark: Sapling bark is thin, smoothish, reddish to gray-brown, and begins peeling in narrow fibers. Mature and old bark... Dormant: Evergreen flattened sprays persist, with no conspicuous scaled winter bud; growth ends in tightly...	Tolerance: Very shade tolerant, able to persist beneath mixed canopies, although successful... Site: Occupies a broad moisture range from swamps and fens to dry cliffs, commonly favored by calcareous or...
Eastern redcedar <i>Juniperus virginiana</i>	Foliage: Adult branchlets carry tiny opposite scale leaves tightly appressed in four ranks, forming rounded to four-... Bark: Young bark is thin reddish brown to gray and begins peeling early. Mature bark shreds in long narrow fibrous... Dormant: Evergreen scale leaves or prickly juvenile leaves persist, and there is no large conspicuous...	Tolerance: Shade intolerant to only weakly tolerant; seedlings establish in open ground and... Site: Extremely site-flexible on dry rocky, sandy, clayey, and calcareous soils, including shallow...

Part III - Fillable instructional sessions

Complete sessions in order when possible. Closed-book retrieval means close or hide the teaching atlas before answering.

Session A - Aspens and balsam poplar

Purpose: organ comparisons across age and clonal form. Taxon scope: listed new taxa. Feedback: during instruction.

Before the session: which species or comparison currently feels least reliable?

Quaking aspen: one diagnostic character plus one silvics anchor

Bigtooth aspen: one diagnostic character plus one silvics anchor

Balsam poplar: one diagnostic character plus one silvics anchor

Speckled alder: one diagnostic character plus one silvics anchor

American mountain-ash: one diagnostic character plus one silvics anchor

Showy mountain-ash: one diagnostic character plus one silvics anchor

Exit retrieval: state the most important correction or strengthened distinction

☐ Session A complete (enter X)

Session B - Populus and transition retrieval

Purpose: unfamiliar isolated specimens and stand-context decisions. Taxon scope: session a taxa plus due boreal associates. Feedback: after batch submission.

Before the session: which species or comparison currently feels least reliable?

1. Silvics

Explain how Gray birch normally regenerates, including establishment requirements and any seed, mast, sprouting, layering, or suckering behavior that materially affects a prescription.

Response

closed-book retrieval

2. Silvics

Give one operational management implication for Norway maple and explain its response to an appropriate silvicultural system or cutting pattern.

Response

closed-book retrieval

3. Forest Association

Place Sugar maple in the course region and in a characteristic forest association. Explain why the context raises or lowers its plausibility without treating habitat as proof of identity.

Response

closed-book retrieval

4. Forest Association

A field crew reports Sweet birch from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response

closed-book retrieval

5. Silvics

Place Balsam fir in succession: give its shade tolerance, explain how it recruits or persists, and characterize its longevity or growth strategy.

Response

closed-book retrieval

6. Silvics

For Showy mountain-ash (*Sorbus decora*), describe the moisture, drainage, and soil setting a forester should expect, then place it in its climatic or forest-association context.

Response

closed-book retrieval

7. Silvics

Explain how American hornbeam normally regenerates, including establishment requirements and any seed, mast, sprouting, layering, or suckering behavior that materially affects a prescription.

Response

closed-book retrieval

8. Silvics

A stand containing Pin cherry is disturbed. Explain the likely biological response and one important damaging agent or vulnerability that changes the prognosis.

Response

closed-book retrieval

9. Forest Association

A field crew reports Red maple from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response

closed-book retrieval

10. Silvics

For Pagoda dogwood (*Cornus alternifolia*), describe the moisture, drainage, and soil setting a forester should expect, then place it in its climatic or forest-association context.

Response

closed-book retrieval

Exit retrieval: state the most important correction or strengthened distinction

☐ Session B complete (enter X)

Session C - Alder and mountain-ash comparisons

Purpose: fruit foliage and growth-form contrasts. Taxon scope: listed new taxa plus cs22. Feedback: after response.

Before the session: which species or comparison currently feels least reliable?

Sugar / black / Norway maple

Course separator: Petiole latex; leaf underside hair and stipules; bud size/shape; samara angle

My visible separator

Best resolving view

Striped / mountain maple

Course separator: Green-white striped bark versus unstriped; leaf size/base; fruit cluster posture; growth form

My visible separator

Best resolving view

Paper / heart-leaved / gray birch

Course separator: Leaf base and vein count; twig resin glands; peeling versus tight bark; black branch marks; elevation

My visible separator

Best resolving view

Yellow / sweet birch

Course separator: Both wintergreen; yellow birch golden curling bark and spur shoots versus sweet birch dark blocky bark

My visible separator

Best resolving view

White / black / red / Norway spruce

Course separator: Cone size and scale margin; twig hairs; bud resin; foliage color/odor; pendulous branchlets

My visible separator

Best resolving view

Spruce / fir / hemlock

Course separator: Woody pegs versus pad base versus petiole; needle cross-section; cone orientation and persistence

My visible separator

Best resolving view

Red / jack / Scots / pitch pine

Course separator: Fascicle count/length; bend test; cone curvature/prickles; upper-bole orange bark; epicormic shoots

My visible separator

Best resolving view

Tamarack / European larch

Course separator: Cone size and scale count/edge; needle cluster density; native peatland versus planting

My visible separator

Best resolving view

Black / pin / chokecherry

Course separator: Fruit cluster architecture; petiole glands; lower-midvein hair; bark age; whole pioneer/clonal form

My visible separator

Best resolving view

American hornbeam / eastern hophornbeam

Course separator: Muscular smooth bole and three-lobed fruit bracts versus shreddy bark and hoplike sacs

My visible separator

Best resolving view

Yellow birch / black cherry bark

Course separator: Golden horizontal curls and wintergreen twig versus burnt-chip plates, petiole glands, and almond twig odor

My visible separator

Best resolving view

Serviceberry / mountain-ashes / chokecherry

Course separator: Simple versus compound leaves; pome calyx versus drupe; leaflet proportions; buds; fruit architecture

My visible separator

Best resolving view

Exit retrieval: state the most important correction or strengthened distinction

☐ Session C complete (enter X)

Session D - Boreal-transition field laboratory

Purpose: regional assemblage and single-organ stations. Taxon scope: all module taxa plus representative conifers and hardwoods. Feedback: after station batch submission.

Before the session: which species or comparison currently feels least reliable?

Practice 1

Identity: common OR scientific name



Evidence asset INAT_PHOTO_614771142 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 2

Identity: common OR scientific name



Evidence asset INAT_PHOTO_631489003 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

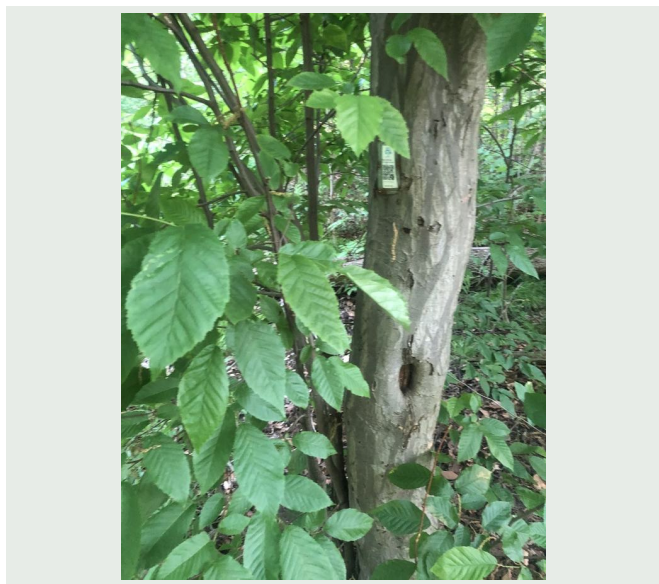
Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 3

Identity: common OR scientific name



Evidence asset INAT_PHOTO_678706139 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 4

Identity: common OR scientific name



Evidence asset INAT_PHOTO_648309432 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 5

Identity: common OR scientific name



Evidence asset INAT_PHOTO_706485056 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 6

Identity: common OR scientific name



Evidence asset INAT_PHOTO_643127949 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 7

Identity: common OR scientific name



Evidence asset INAT_PHOTO_651017782 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

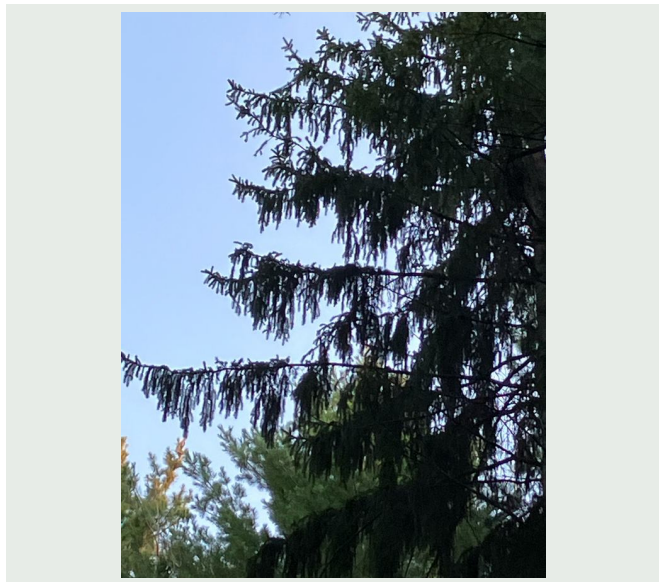
Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 8

Identity: common OR scientific name



Evidence asset INAT_PHOTO_711263168 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 9

Identity: common OR scientific name



Evidence asset INAT_PHOTO_709279670 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 10

Identity: common OR scientific name



Evidence asset INAT_PHOTO_298582541 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Exit retrieval: state the most important correction or strengthened distinction

☐ Session D complete (enter X)

Session E - First substantial cumulative review

Purpose: formal-practical preparation without image reuse. Taxon scope: cr 01 eligible taxa prioritized by mastery. Feedback: after batch submission.

Before the session: which species or comparison currently feels least reliable?

1. Silvics

Predict the role of Pagoda dogwood from suppression through canopy recruitment. Tie the prediction to shade tolerance, successional status, and growth or longevity.

Response

closed-book retrieval

2. Silvics

A stand containing American hornbeam is disturbed. Explain the likely biological response and one important damaging agent or vulnerability that changes the prognosis.

Response

closed-book retrieval

3. Silvics

For American mountain-ash (*Sorbus americana*), describe the moisture, drainage, and soil setting a forester should expect, then place it in its climatic or forest-association context.

Response

closed-book retrieval

4. Silvics

A stand prescription lists Pagoda dogwood as an objective species. State the site conditions that favor it and one site or climatic constraint that would make that objective questionable.

Response

closed-book retrieval

5. Silvics

Predict the role of Black cherry from suppression through canopy recruitment. Tie the prediction to shade tolerance, successional status, and growth or longevity.

Response

closed-book retrieval

6. Silvics

For Balsam poplar (*Populus balsamifera*), describe the moisture, drainage, and soil setting a forester should expect, then place it in its climatic or forest-association context.

Response

closed-book retrieval

7. Nomenclature

Give the accepted scientific name for Sweet birch.

Response

closed-book retrieval

8. Silvics

After a harvest, what regeneration pathway should a forester expect from American beech, and what substrate, light, moisture, or advance-regeneration conditions control success?

Response

closed-book retrieval

9. Silvics

Predict the role of Black maple from suppression through canopy recruitment. Tie the prediction to shade tolerance, successional status, and growth or longevity.

Response

closed-book retrieval

10. Silvics

A stand prescription lists Eastern redcedar as an objective species. State the site conditions that favor it and one site or climatic constraint that would make that objective questionable.

Response

closed-book retrieval

Exit retrieval: state the most important correction or strengthened distinction

☐ Session E complete (enter X)

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Species-profile references

Quaking Aspen - Silvics of North America: <https://research.fs.usda.gov/silvics/quaking-aspen> - Range, sites, associates, seed and clonal regeneration, succession, disturbance response, growth, damaging agents, and management.

Populus tremuloides - quaking poplar: <https://gobotany.nativeplanttrust.org/species/populus/tremuloides/> - Regional morphology, petiole orientation, leaf measurements, tooth counts, habitat, hybrids, and defensible comparison with bigtooth aspen.

Populus tremuloides, quaking aspen - Fire Effects Information System: <https://research.fs.usda.gov/feis/species-reviews/poptre> - Fire ecology, postfire suckering, plant communities, browse, successional behavior, and management evidence.

Bigtooth Aspen - Silvics of North America: <https://research.fs.usda.gov/silvics/bigtooth-aspen> - Distribution, soils and climate, associates, seed and sucker reproduction, succession, disturbance, growth, damaging agents, and management.

Populus grandidentata - big-toothed poplar: <https://gobotany.nativeplanttrust.org/species/populus/grandidentata/> - Regional leaf measurements and tooth counts, bud pubescence, habitat, taxonomy, hybrids, and comparison with quaking aspen.

Populus grandidentata, bigtooth aspen - Fire Effects Information System: <https://research.fs.usda.gov/feis/species-reviews/popgra> - Plant communities, clonal growth, postfire response, succession, wildlife use, and management evidence.

Balsam Poplar - Silvics of North America: <https://research.fs.usda.gov/silvics/balsam-poplar> - Range, floodplain and upland sites, associates, sexual and vegetative reproduction, succession, growth, wildlife use, and management.

Populus balsamifera - balsam poplar: <https://gobotany.nativeplanttrust.org/species/populus/balsamifera/> - Regional morphology, resin characters, petiole and twig traits, habitat, taxonomy, and hybrid or cottonwood comparisons.

Populus - key for mature vegetative material: <https://gobotany.nativeplanttrust.org/dkey/populus-key-for-mature-vegetative-material/> - Comparative long-shoot foliage, petiole compression, glands, branchlet form, and winter-bud position across regional poplars.

Alnus incana - speckled alder: <https://gobotany.nativeplanttrust.org/species/alnus/incana/> - Regional taxonomy, wetland status, leaves, bark lenticels, buds, reproductive structures, hybrids, and comparisons with smooth and green alder.

Alnus - dichotomous key: <https://gobotany.nativeplanttrust.org/dkey/alnus/> - Direct diagnostic contrasts in bud stalks and scales, leaf serration and bloom, bark lenticels, resin, nutlet wings, and inflorescence architecture.

Speckled Alder - USDA NRCS Plant Guide: https://plants.usda.gov/DocumentLibrary/plantguide/pdf/cs_alinr.pdf - Accepted infraspecific name, distribution, nitrogen fixation, wet-site adaptation, reproduction, wildlife use, restoration functions, and management.

Sorbus americana - American mountain-ash: <https://gobotany.nativeplanttrust.org/species/sorbus/americana/> - Regional morphology, leaf and bud characters, fruit, habitat, taxonomy, and direct comparison with showy mountain-ash and European rowan.

Sorbus americana, American mountain-ash - Fire Effects Information System: <https://research.fs.usda.gov/feis/species-reviews/sorame> - Distribution, sites, communities, regeneration, shade tolerance, succession, fire response, browse, wildlife use, and management.

Sorbus seed biology - Woody Plant Seed Manual: <https://www.fs.usda.gov/nsi/Wpsm/Sorbus.pdf> - Genus-level flowering, fruiting, seed collection, dormancy, stratification, germination, and propagation evidence.

Sorbus decora - showy mountain-ash: <https://gobotany.nativeplanttrust.org/species/sorbus/decora/> - Regional morphology, taxonomic variation, flowering time, leaflet and bud characters, fruit measurements, habitat, and comparison with American mountain-ash.

Showy mountain-ash - Ontario Tree Atlas: <https://www.ontario.ca/page/showy-mountain-ash> - Ontario distribution, form, bark, leaflet number and size, flowers, fruit persistence, moisture, soils, shade, and wildlife value.

Native Trees and Shrubs for Maine Landscapes: Showy Mountain Ash: <https://extension.umaine.edu/publications/2582e/> - Cold-climate landscape performance, habit, site requirements, flowers, fruit, disease considerations, and regional management context.